Algebra 1
Unit 7
Lesson 1 Practice

Name____Answer key______
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For questions 1-2, consider the expression $16y^3 - 8y^2 - 20y$.

1. Write an equivalent expression that uses the greatest common monomial factor.

$$4y\,(4y^2 - 2y - 5)$$

2. Write an equivalent expression that does not use the greatest common monomial factor. Answers will vary. Some options:

$2\,(8y^3 - 4y^2 - 10y)$ $\quad 2y\,(8y^2 - 4y - 10)$
$y\,(16y^2 - 8y - 20)$

For questions 3-6, write an equivalent expression for the given expression that uses the greatest common monomial factor.

3. $6f^5g^3 + 12f^2g - 30f^3g^3$

$6f^2g\,(f^3g^2 + 2 - 5fg^2)$

4. $55s^4h^2 + 121s^6h^2 + 11s^2h$

$11s^2h\,(5s^2h + 11s^4h + 1)$

5. $80db - 20d^7b^3$

$20db(4 - d^6b^2)$

6. $81x^5y^8 - 18x^2y^3 - 90x^6y^6$

$9x^2y^3\,(9x^3y^5 - 2 - 10x^4y^3)$

For questions 7-10, find the greatest common monomial factor of the expression.

7. $21a^3b^4 - 49ab$

$7ab$

8. $24qc^2 + 20q^5c + 32q^7c^2$

$4qc$

9. $16r^2m^8 - 64r^8m^2 + 40r^3m^3$

$8r^2m^2$

10. $6x^6y^5 - 21x^7y^8 - 36x^2y^4$

$3x^2y^4$

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For questions 1-4, consider the trinomial $x^2 - 10x + 21$.

1. How many ^{+}x tiles are needed for an algebra tiles diagram of the trinomial?

   0

2. How many ^{-}x tiles are needed for an algebra tiles diagram of the trinomial?

   10

3. Draw a diagram of the algebra tiles that show the trinomial $x^2 - 10x + 21$.

x^2	$-x$	$-x$	$-x$	$-x$	$-x$	$-x$	$-x$
$-x$	$+1$	$+1$	$+1$	$+1$	$+1$	$+1$	$+1$
$-x$	$+1$	$+1$	$+1$	$+1$	$+1$	$+1$	$+1$
$-x$	$+1$	$+1$	$+1$	$+1$	$+1$	$+1$	$+1$

4. Complete the table for $x^2 - 10x + 21$.

	x	$\underline{-7}$
x	x^2	$\underline{-7x}$
$\underline{-3}$	$\underline{-3x}$	21

For questions 5-8, consider the trinomial $x^2 - 4x - 12$.

5. How many ^{+}x tiles are needed for an algebra tiles diagram of the trinomial?
   2

6. How many ^{-}x tiles are needed for an algebra tiles diagram of the trinomial?
   6

7. Draw a diagram of the algebra tiles that show the trinomial $x^2 - 4x - 12$.

x^2	$-x$	$-x$	$-x$	$-x$	$-x$	$-x$
$+x$	-1	-1	-1	-1	-1	-1
$+x$	-1	-1	-1	-1	-1	-1

8. Complete the table for $x^2 - 4x - 12$.

	x	$+2$
x	x^2	$2x$
-6	$-6x$	-12

9. Complete the table for $x^2 + x - 110$.

	x	$+11$
x	x^2	$11x$
-10	$-10x$	-110

10. Complete the table for $x^2 + 3x - 40$.

	x	$+8$
x	x^2	$8x$
-5	$-5x$	-40

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For questions 1-5, factor each quadratic expression using the method of your choice.

1. $b^2 + 17b + 72$

$(b + 9)(b + 8)$

2. $h^2 - 23h + 132$

$(h - 11)(h - 12)$

3. $x^2 + 15x + 56$

$(x + 7)(x + 8)$

4. $y^2 - 3y - 18$

$(y - 6)(y + 3)$

5. $x^2 - 7x - 30$

$(x - 10)(x + 3)$

For questions 6-7, consider the trinomial $f^2 - 10f + 9$.

6. Complete the area model. $(f - 9)(f - 1)$

	f	-9
f	f^2	$-9f$
-1	$-f$	9

7. What are the two binomial factors for $f^2 - 10f + 9$?

$(f - 9)(f - 1)$

For questions 8-9, consider the trinomial $m^2 - 21m + 108$.

8. Complete the area model.

	m	-12
m	m^2	$-12m$
-9	$-9m$	$+108$

9. What are the two binomial factors for $m^2 - 21m + 108$?

$(m - 12)(m - 9)$

10. Factor $x^2 - 12x + 35$ using factoring by grouping.

$x^2 - 7x - 5x + 35$
$x(x - 7) - 5(x - 7)$
$(x - 5)(x - 7)$

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For questions 1-5, factor each quadratic expression using the method of your choice.

1. $5b^2 + 46b + 48$

$5b^2 + 40b + 6b + 48$
$5b\,(b + 8) + 6(b + 8)$
$(5b + 6)\,(b + 8)$

2. $4a^2 + 27a + 18$

$4a^2 + 24a + 3a + 18$
$4a\,(a + 6) + 3(a + 6)$
$(4a + 3)\,(a + 6)$

3. $2x^2 + 3x - 27$

$2x^2 + 9x - 6x - 27$
$x\,(2x + 9) - 3\,(2x + 9)$
$(x - 3)\,(2x + 9)$

4. $3k^2 + 5k - 28$

$3k^2 + 12k - 7k - 28$
$3k\,(k + 4) - 7\,(k + 4)$
$(3k - 7)\;(k + 4)$

5. $5z^2 - 28z - 96$

$5z^2 - 40z + 12z - 96$
$5z(z - 8) + 12\,(z - 8)$
$(5z + 12)(z - 8)$

For questions 6-7, consider the trinomial $4x^2 - 41x + 72$.

6. Complete the area model.

	$4x$	-9
x	$4x^2$	$-9x$
-8	$-32x$	72

7. What are the two binomial factors for $4x^2 - 41x + 72$?

$(4x - 9)(x - 8)$

For questions 8-9, consider the trinomial $2m^2 - 9m - 110$.

8. Complete the area model.

	$2m$	$+11$
m	$2m^2$	$11m$
-10	$-20m$	-110

9. What are the two binomial factors for $2m^2 - 9m - 110$?

$(2m + 11)(m - 10)$

10. Factor $3x^2 - 4x - 32$ using factoring by grouping.

$3x^2 - 12x + 8x - 32$
$3x\ (x - 4) + 8\ (x - 4)$
$(3x + 8)(x - 4)$

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For questions 1-3, factor each quadratic expression using the method of your choice.

1. $10b^2 - 9b - 9$

$10b^2 - 15b + 6b - 9$
$5b(2b - 3) + 3\,(2b - 3)$
$(5b + 3)(2b - 3)$

2. $12x^2 + 19x - 21$

$12x^2 + 28x - 9x - 21$
$4x(3x + 7) - 3(3x + 7)$
$(4x - 3)\,(3x + 7)$

3. $6d^2 - 13d + 6$

$6d^2 - 9d - 4d + 6$
$3d(2d - 3) - 2\,(2d - 3)$
$(3d - 2)\,(2d - 3)$

For questions 4-5, consider the trinomial $10x^2 - 61x + 72$.

4. Write an expanded form of the trinomial that can be used to factor the expression by grouping.

$10x^2 - 45x - 16x + 72$

5. Use the expanded form to factor the expression by grouping.

$5x(2x - 9) - 8(2x - 9)$
$(5x - 8)\,(2x - 9)$

For questions 6-7, consider the trinomial $8x^2 + 46x + 45$.

6. Complete the area model.

	$4x$	5
$2x$	$8x^2$	$10x$
9	$36x$	45

7. What are the two binomial factors for $8x^2 + 46x + 45$?

$(4x + 5)\ (2x + 9)$

For questions 8-9, consider the trinomial $10t^2 + 9\mathrm{t} - 9$.

8. Complete the area model.

	$5t$	-3
$2t$	$10t^2$	$-6t$
3	$15t$	-9

9. What are the two binomial factors for $10t^2 + 9t - 9$?

$(5t - 3)(2t + 3)$

10. Factor $6x^2 + 57x + 132$ using factoring by grouping.

$3(2x^2 + 19x + 44)$
$3(2x^2 + 11x + 8x + 44)$
$3\big(x\ (2x + 11) + 4\ (2x + 11)\big)$
$3(x + 4)(2x + 11)$

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For questions 1-6, factor each expression by grouping.

1. $2y^4 - 8y^3 + 10y^2 - 40y$

$2y^3(y-4) + 10y(y-4)$
$(2y^3 + 10y)(y-4)$
$2y(y^2+5)(y-4)$

2. $15x^2 + 62x + 63$

$15x^2 + 35x + 27x + 63$
$5x(3x+7) + 9(3x+7)$
$(5x+9)(3x+7)$

3. $-12a + 20ab - 35b + 21$

$4a(-3+5b) - 7(5b-3)$
$(4a-7)(5b-3)$

4. $6x^2 - x - 77$

$6x^2 - 22x + 21x - 77$
$2x(3x-11) + 7(3x-11)$
$(2x+7)(3x-11)$

5. $3x^2 + 12x + 6x + 24$

$3x(x+4) + 6(x+4)$
$(3x+6)(x+4)$
$3(x+2)(x+4)$

6. $4x^2 + 38x + 18$

$2\,(2x^2 + 19x + 9)$
$2\,(2x^2 + 18x + x + 9)$
$2\,\left(2x(x+9) + 1(x+9)\right)$
$2\,(2x+1)(x+9)$

For questions 7-10, consider the trinomial $15x^2 - 34x - 16$.

7. List all the factor pairs for -240.

$-1,240$	$-6,40$	$1,-240$	$6,-40$
$-2,120$	$-8,30$	$2,-120$	$8,-30$
$-3,80$	$-10,24$	$3,-80$	$10,-24$
$-4,60$	$-12,20$	$4,-60$	$12,-20$
$-5,48$	$-15,16$	$5,-48$	$15,-16$

8. Find the factor pair for -240 that add to the sum of -34.

$-40,6$

9. Use question 8 to write the expanded form of $15x^2 - 34x - 16$ that can be used to factor the expression by grouping.

$15x^2 - 40x + 6x - 16$

10. Factor the expression $15x^2 - 34x - 16$ by grouping.

$15x^2 - 40x + 6x - 16$
$5x(3x - 8) + 2(3x - 8)$
$(5x + 2)(3x - 8)$

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For questions 1-2, consider the perfect square trinomial $25x^2 + 30x + 9$.

1. Rewrite the trinomial in the form $a^2 + 2ab + b^2$.

   $(5x)^2 + 2(5x)(3) + (3)^2$

2. Factor the perfect square trinomial.

   $(5x + 3)^2$

For questions 3-4, consider the perfect square trinomial $9x^2 - 42x + 49$.

3. Rewrite the trinomial in the form $a^2 - 2ab + b^2$.

   $(3x)^2 - 2(3x)(7) + (7)^2$

4. Factor the perfect square trinomial.

   $(3x - 7)^2$

For questions 5-8, determine whether the quadratic expression is a perfect square trinomial. Then, factor the expression.

5. $w^2 - 24w + 144$

Yes.
$(w - 12)^2$

6. $p^2 + 2pq + q^2$

Yes.
$(p + q)^2$

7. $49v^2 - 25$

No.
$(7v - 5)(7v + 5)$

8. $16y^2 - 256y + 32$

No.
$16(y^2 - 16y + 2)$

For questions 9-10, factor the trinomial.

9. $36a^2 - 60a + 25$

$(6a - 5)^2$

10. $p^2 - 18p + 81$

$(p - 9)^2$

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For questions 1-2, consider the expression $36x^2 - 64y^2$.

1. Rewrite the expression as a difference of two squares.

$(6x)^2 - (8y)^2$

2. Factor the expression.

$(6x + 8y)(6x - 8y)$

For questions 3-4, consider the expression $121m^2 - 25n^2$.

3. Rewrite the expression as a difference of two squares.

$(11m)^2 - (5n)^2$

4. Factor the expression.

$(11m - 5n)(11m + 5n)$

For questions 5-8, determine whether the quadratic expression is a difference of two squares. Then, factor the expression.

5. $49s^2 - 144m^2$

Yes
$(7s + 12m)(7s - 12m)$

6. $49g^2 + 84g + 36$

No
$(7g + 6)^2$

7. $9 - 4r^2$

Yes
$(2r + 3)(-2r + 3)$

8. $-100 + w^2$

Yes
$(w + 10)(w - 10)$

For questions 9-10, factor the expression.

9. $x^2 - 4x + 4$

$(x - 2)^2$

10. $81v^2 - 9p^2$

$9\,(9v^2 - p^2)$
$9\,(3v - p)(3v + p)$